ALI:

Public key: (7, 33)

Private key: (3,33)

Arham:

Public key: (5, 221)

Private key: (77, 221)

Message (M) = 3

At Ali's end:

Signing => $(M)^{PK(Ali)}$ mod n

Encryption => $((M)^{PK(Ali)})^{PBK(Arham)}$ mod n

Signing = $(3)^3$ mod 33 = 27

Encryption = $(27)^5$ mod 221 =(MOD(POWER(27,5),221)) = 40

At Arham's end:

C = 40

Decrypt => $(40)^{77}$ mod 221 =
MOD((MOD((92*92*92*92),221)*MOD((92*92*92*92),221)*MOD((92*92*92*92),221)*MOD((92*92),221)*MOD(POWER(40,7),221)),221) = 27

Verifying = $(27)^7$ mod 33 =MOD(POWER(27,7),33) = 3

Hence verified!